

쥐 모델에서 전극 배치에 따른 Temporal Interference 전기장 분포 특성 분석

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Effect of Electrode Configuration on Temporal Interference Electric Fields in a Rat Model

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Abstract

Temporal interference stimulation (TIS) is a promising non-invasive neuromodulation technique capable of targeting deep tissue regions through the interaction of high-frequency electric fields, with its effectiveness strongly dependent on electrode configuration. In this study, we investigated the effect of electrode placement on electric field distribution in a two-dimensional rat model using finite element simulations. A realistic anatomical model incorporating multiple tissue types was developed, and TIS was applied using dual high-frequency signals (8 kHz and 8.01 kHz) with an input voltage of 3 V per channel across multiple configurations in transverse and sagittal planes. The results showed that electrode configuration significantly affects the electric potential within the bladder region, with transverse configurations producing relatively uniform distributions (1.46–1.61 V) and sagittal configurations exhibiting a wider range (1.00–2.37 V). Configurations with the active electrode positioned closer to the bladder and the return electrode placed farther away resulted in higher field concentration and improved targeting, whereas current pathways passing near low-conductivity regions such as bone reduced the effective field intensity. These findings highlight the importance of optimized electrode placement and tissue-aware design for enhancing stimulation efficiency.

1. Background

Underactive bladder (UAB) is a neurological condition characterized by impaired bladder contraction, leading to incomplete emptying and reduced quality of life. Current treatment options, including pharmacological therapy and catheterization, present limitations such as low efficacy and patient discomfort.

Recently, non-invasive neuromodulation techniques have gained attention as alternative therapeutic approaches. Among these, temporal interference stimulation (TIS) enables the generation of low-frequency stimulation effects in deep tissues by applying multiple high-frequency electric fields [1]. This approach has the potential to target deep neural structures without directly stimulating superficial tissues and has been demonstrated in recent human studies [3].

However, the effectiveness of TIS is highly dependent on electrode configuration, which determines the spatial distribution and focality of the induced electric field [4]. Therefore, systematic analysis of electrode placement is essential to optimize stimulation of the bladder region [5]. In this study, we investigate the effect of electrode configuration on the electric field distribution in a rat model using computational simulations.

2. Method

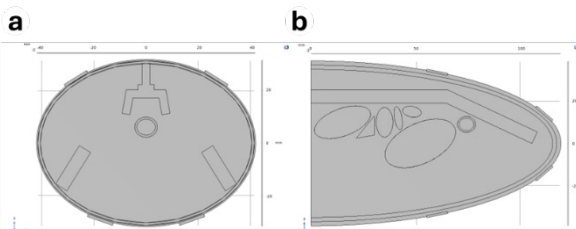


Fig. 1. (a) Transverse plane model. (b) Sagittal plane model

A two-dimensional rat model was developed to approximate realistic anatomical structures, including skin, fat, bone, and internal organs such as the bladder, liver, intestine, kidney, and stomach. The model also included surrounding air and urine regions, and electrical properties were assigned to each tissue domain based on standard

bioelectrical characteristics. Two cross-sectional planes were considered for analysis: a transverse plane and a sagittal plane, as illustrated in Fig. 1.

Multiple electrode configurations were designed by varying electrode placement on the body surface. Square electrodes (1 cm × 1 cm) with a conductive gel layer were used to improve electrode–skin contact. Each stimulation channel was modeled with an independent ground. Temporal interference stimulation was implemented using two high-frequency sinusoidal signals at 8 kHz and 8.01 kHz, with 3 V applied to each channel, generating a low-frequency envelope within the tissue.

Simulations were performed using COMSOL Multiphysics under steady-state conditions. The electric potential distribution was calculated for each configuration under identical boundary conditions. The mean electric potential within the bladder region was extracted as a quantitative metric to evaluate the effect of electrode placement on stimulation intensity.

3. Result

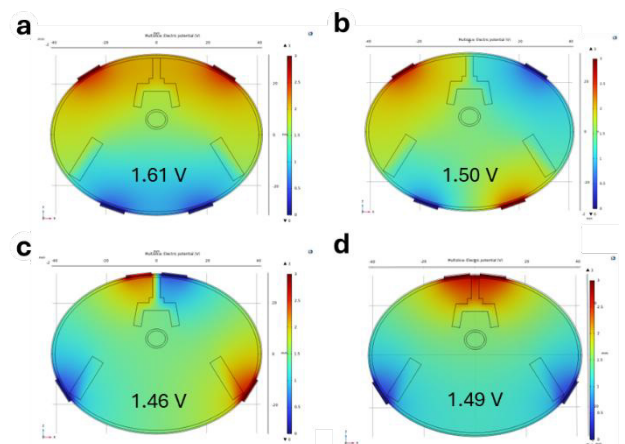


Fig. 2. Electric potential distribution in the transverse plane for four representative electrode configurations (a–d). The values indicate the mean electric potential within the bladder region.

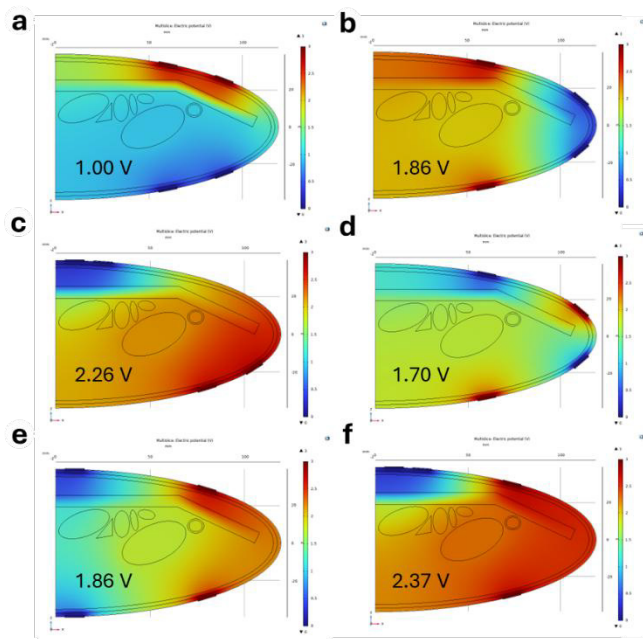


Fig. 3. Electric potential distribution in the sagittal plane for six representative electrode configurations (a–f). The values indicate the mean electric potential within the bladder region.

The electric potential distribution varied significantly depending on electrode configuration in both transverse and sagittal planes, demonstrating the strong influence of electrode placement on field characteristics.

In the transverse plane (Fig. 2), the electric potential within the bladder region showed relatively limited variation across configurations, with mean values ranging from 1.46 V to 1.61 V. Configuration (a) produced the highest bladder potential (1.61 V), while configuration (c) exhibited the lowest value (1.46 V). The overall field distribution in this plane remained relatively symmetric, indicating that transverse electrode arrangements result in more uniform but less focal stimulation patterns.

In contrast, the sagittal plane (Fig. 3) exhibited a substantially greater variation in electric potential, with values ranging from 1.00 V to 2.37 V. Configuration (f) achieved the highest bladder potential (2.37 V), followed by configurations (c) (2.26 V) and (b)/(e) (1.86 V), whereas configuration (a) resulted in the lowest value (1.00 V). These results highlight that sagittal electrode configurations have a stronger influence on both field magnitude and spatial targeting.

A key observation is that configurations with the active (driving) electrode positioned closer to the bladder region and the return (ground) electrode located farther away consistently produced higher electric potential within the target region. In these cases, the electric field lines were more effectively directed and concentrated toward the bladder, enhancing both field intensity and focality. Conversely, when the return electrode was placed closer to the bladder, the field distribution became more dispersed, reducing stimulation efficiency.

Additionally, configurations in which the current path passed through or near low-conductivity regions such as bone tended to exhibit reduced electric potential within the bladder. This suggests that tissue heterogeneity, particularly the presence of resistive structures, can significantly attenuate field propagation and diminish effective stimulation at the target site.

Overall, these findings indicate that both electrode proximity to the target region and the spatial separation between active and return electrodes, as well as underlying tissue composition, play critical roles in shaping the electric field distribution. Proper electrode arrangement can significantly enhance targeted stimulation while minimizing undesired spread to surrounding tissues.

5. Conclusion

This study demonstrates that electrode configuration is a critical factor in determining the effectiveness of temporal interference stimulation for bladder-targeted neuromodulation. While transverse configurations produced relatively uniform field distributions with limited variation in intensity, sagittal configurations enabled significantly greater modulation of electric potential within the bladder region. In particular, positioning the active electrode closer to the target while maintaining sufficient distance from the return electrode enhanced field concentration and improved stimulation efficiency.

6. Acknowledgement

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7. Reference

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